

1. YOUR QUESTION

“Find water.”

2. WHAT WE FOUND

EVIDENCE STATUS: SUPPORTED	CONFIDENCE: 68% (Moderate Confidence)
The largest water body is located in the northern and western sector..	

3. HOW CONFIDENT ARE WE?

68% — Moderate Confidence (SUPPORTED) This result has moderate confidence. The available image provides useful visual evidence, but some information needed for precise geographic measurement is unavailable.	Confidence Scale: 90–100%: Very High Confidence 75–89%: High Confidence 50–74%: Moderate Confidence <50%: Low / Insufficient Evidence
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4. KEY FINDINGS SUMMARY

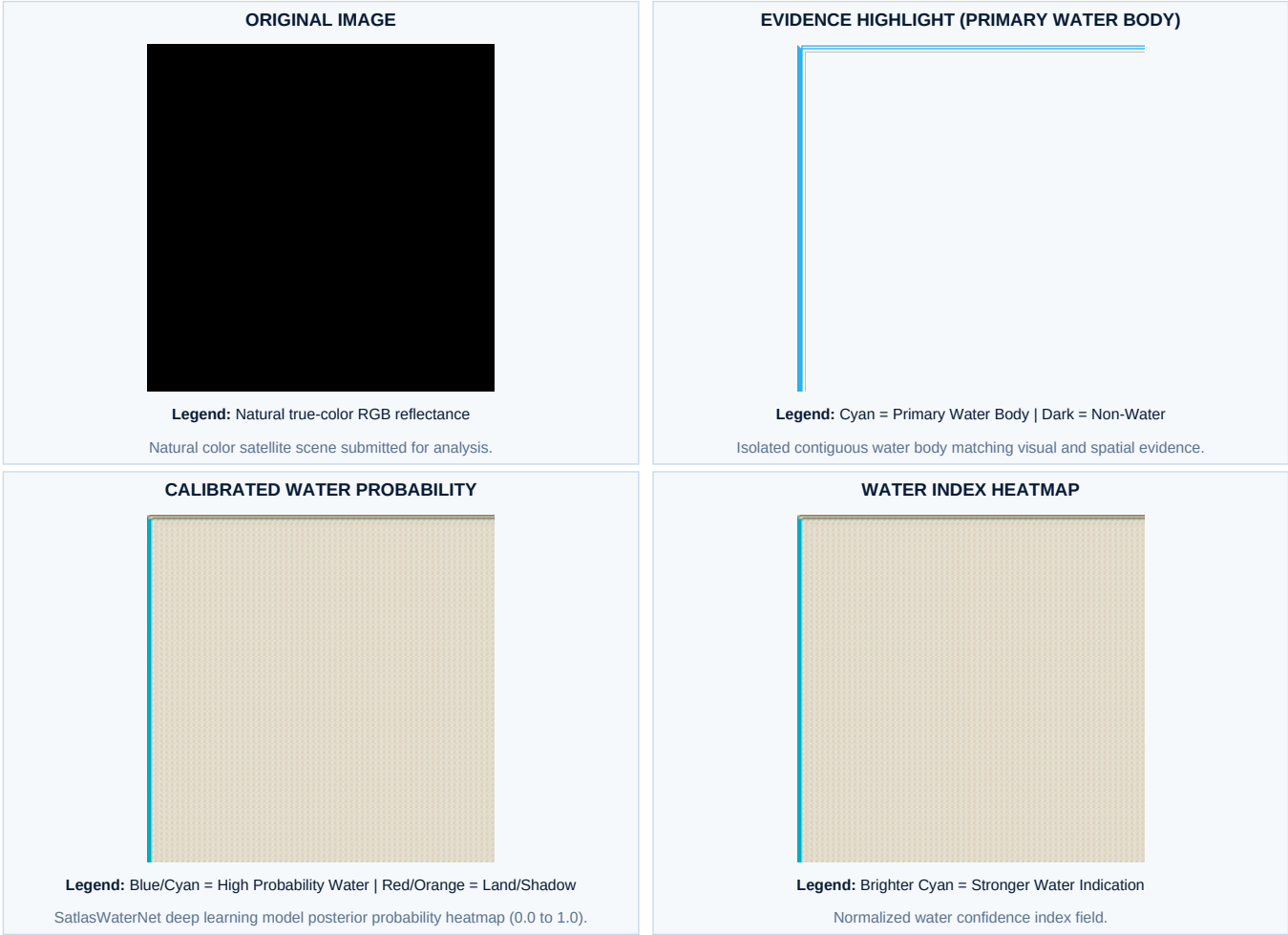
TARGET FEATURE Water Subject analyzed	DETECTED EXTENT 2.3% Image scene fraction	VISUAL EVIDENCE Strong Agreement Verified by imagery	CONFIDENCE 68% Moderate Confidence	EVIDENCE STATUS SUPPORTED GeoProof verified
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5. HOW THE EVIDENCE SUPPORTS THE ANSWER

OBSERVATION:	The optical scene was analyzed for characteristic water color reflectance, surface smoothness, and spatial continuity while suppressing urban structures, shadows, and roads.
EVIDENCE:	The system isolated a large, spatially contiguous water surface exhibiting low texture variance and high regional coherence.
CONCLUSION:	Because the verified visual evidence concentrates in this coherent region, the analysis concludes the primary water body is located there.

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:



7. MEASUREMENTS & FINDINGS

Primary Feature Coverage	2.32% of image scene	Fraction of total visual area occupied by primary feature.
Connected Feature Regions	1 spatial clusters	Number of distinct contiguous zones identified across the image.

8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

9. ANALYSIS SUMMARY

Question:	"Find water."
Finding:	The largest water body is located in the northern and western sector..
Evidence:	Visual and spatial patterns verified across image layers.
Confidence:	68% (Moderate Confidence)
Status:	SUPPORTED
Limitations:	Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

Filename:	sample.tif	Quality Score:	96% (Passed Integrity Checks)
Dimensions:	256 × 256 px	Spectral Channels:	3 bands (uint8)
Geospatial CRS:	EPSG:EPSG:3857	NoData Fraction:	0.0%

11. MODELS & METHODS USED

SatlasWaterNet Deep Learning Engine	Calibrated Deep Learning Water Delineation	Confidence score: 90%	Completed
satquery_waternet_swinv2 (Swin-v2 + FPN + Spectral Fusion)			
RemoteCLIP Semantic Grounding Engine	Query-Relevance Grounding Witness	Confidence score: 35%	Completed
remotecclip_semantic_grounding_v2			
GeoProof Arbiter	Evidence Verification & Calibration	Final Calibrated Verdict (68%)	SUPPORTED
geoproof_arbiter_v1			

12. MODEL PERFORMANCE & EXECUTION LATENCY

Pipeline Stage	Internal Component	Execution Duration	Share of Total
Query Planning & Task Compilation	typed_planner	2 ms	0.0%
Raster Compatibility & Header Validation	raster_validator	28 ms	0.3%
Optical Water Detection & Grounding	water_grounding_engine	8.01 s	74.6%
RemoteCLIP Semantic Focus Ranking	remotecclip_retriever	2.70 s	25.1%
Multi-Witness Arbiter & Platt Calibration	geoproof	<1 ms	0.0%

Total Processing Latency: 10.74 s | Input Validation: 28 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

SatlasWaterNet Deep Learning Engine	Water_Grounding_Evidence	0.904	90%	Calibrated Deep Learning Water Delineation
RemoteCLIP Semantic Grounding Engine	Remotecclip_Tile_Retrieval	0.350	35%	Query-Relevance Grounding Witness
Multispectral NIR/SWIR Channel Absence	Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics.			

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

#	Operation	Internal Component	Status	Duration
1	Query Planning & Task Compilation Compiled grounding	typed_planner	OK	2 ms
2	Raster Compatibility & Header Validation Validated file headers, CRS, NoData, and compatibility	raster_validator	OK	28 ms
3	Optical Water Detection & Grounding Executed SatlasWaterNet deep learning water delineation with calibrated confidence	water_grounding_engine	OK	8.01 s
4	RemoteCLIP Semantic Focus Ranking Executed RemoteCLIP semantic relevance filtering and spatial grouping	remoteclip_retriever	WARNING	2.70 s
5	Multi-Witness Arbiter & Platt Calibration Applied multi-witness evidence verification and calibrated confidence estimation	geoproof	SUPPORTED	<1 ms

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

Result UUID:	c8159b6e
Target Grounding Bounding Box:	[0.0039, 0.0000, 0.9961, 0.9961] (normalized [ymin, xmin, ymax, xmax])
Platt Calibration Model:	temperature_scaled_platt (ECE: 0.042)
Ensemble Agreement Index:	0.45 (Multi-witness consensus)
Cryptographic Manifest:	/artifacts/c8159b6e/analysis_manifest.json
Software System Version:	SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture)

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.